
Hierarchical Emotion Classification: Evaluating Model Architectures for Fine-Grained Sentiment Analysis

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Abstract

Fine-grained emotion classification requires distinguishing subtle emotional states, yet it is unclear whether explicitly modeling hierarchical structure improves performance over flat approaches. We compare three architectures on the GoEmotions dataset (28 classes): a flat DistilBERT classifier, a hierarchical multi-task model, and a mixture-of-experts (MoE) model. All three achieve similar results, with the flat model performing slightly best (57.7% accuracy, 0.443 macro F1), followed by MoE (57.5%, 0.442) and hierarchical (56.7%, 0.437). The hierarchical model exhibits worse calibration (ECE 0.117 vs. 0.057), while MoE shows intermediate calibration (0.066). Representation analysis reveals that flat and hierarchical models cluster by sentiment without explicit supervision, while MoE experts fail to develop meaningful specialization despite learned routing. Overall, explicitly encoding hierarchy provides limited benefit, suggesting pre-trained transformers already capture emotion structure implicitly.

1 Introduction

Emotion classification is fundamental to NLP with applications from mental health monitoring to customer service automation. While binary sentiment analysis has achieved impressive performance, fine-grained emotion classification remains challenging due to subtle distinctions between related states like "annoyance" and "anger."

Emotions naturally follow hierarchical structure: they can be grouped into positive, negative, and ambiguous sentiments, then refined into basic categories (joy, sadness, anger), which decompose into fine-grained states. This raises an important question: should we explicitly encode this structure into models, or can neural networks learn it implicitly from data?

We investigate whether hierarchical modeling improves fine-grained emotion classification by comparing three architectures on GoEmotions: (1) a flat classifier predicting 28 emotions directly, (2) a hierarchical classifier jointly predicting sentiment and emotions via multi-task learning, and (3) a mixture-of-experts model combining hierarchical prediction with specialized networks. We evaluate predictive performance, calibration, confusion patterns, and learned representations.

This research addresses two questions: Can hierarchical modeling improve fine-grained emotion classification? Is a full mixture-of-experts architecture necessary, or can simpler models capture the benefits of explicit structure? We provide empirical insights into trade-offs between model complexity and performance for practitioners designing emotion classification systems.

2 Background

2.1 Emotion Classification and the GoEmotions Dataset

The GoEmotions dataset contains over 58,000 Reddit comments annotated with 28 emotion categories, making it one of the largest resources for fine-grained emotion classification. The 28 emotions span from *admiration* and *gratitude* to *anger* and *disgust*, and exhibit a natural hierarchical structure: emotions group into three sentiment categories (positive, negative, ambiguous) based on valence, aligning with established psychological frameworks.

2.2 Hierarchical Classification and Mixture of Experts

Hierarchical classification leverages label relationships to improve learning. Two common approaches are: (1) **multi-task learning**, where models jointly predict sentiment and emotions with combined loss

$$\lambda_1 L_{\text{sentiment}} + \lambda_2 L_{\text{emotion}}$$

and (2) **cascaded classification**, which predicts coarse labels first but risks error propagation.

Mixture-of-Experts (MoE) architectures combine multiple expert networks via dynamic gating:

$$\text{output} = \sum_{i=1}^N g_i(x) E_i(x)$$

where $g_i(x)$ denotes the gating weight for expert i . Whether this added complexity yields benefits for emotion classification remains an open question.

2.3 Model Calibration

A model is well-calibrated if predicted probabilities reflect empirical correctness. Expected Calibration Error (ECE) measures average discrepancy between confidence and accuracy across bins, while Maximum Calibration Error (MCE) captures worst-case deviation. Deep networks are often overconfident, making calibration critical for deployment.

3 Methodology

3.1 Dataset and Training

We use the GoEmotions dataset (54,263 Reddit comments, 28 emotions) with splits of 43,410/5,426/5,427 for train/validation/test. We simplify to single-label classification by selecting the first label per example. The 28 emotions map to three sentiment categories: positive (12 emotions), negative (11 emotions), and ambiguous (5 emotions). Inputs are tokenized using DistilBERT with max length 128.

All models train for 5 epochs using AdamW (lr= 2×10^{-5} , weight decay=0.01) with batch sizes 16 (flat/hierarchical) or 24 (MoE). We apply gradient clipping (max norm 1.0) and select best checkpoints based on validation loss or emotion F1.

3.2 Model Architectures

3.2.1 Flat Classifier

DistilBERT-base-uncased encoder with linear head mapping pooled [CLS] (768-dim) to 28 emotion logits, trained with cross-entropy:

$$L_{\text{flat}} = - \sum_{i=1}^{28} y_i \log(\hat{y}_i).$$

3.2.2 Hierarchical Classifier

Shared DistilBERT encoder feeds two heads: a sentiment head (3 classes) and emotion head (28 classes). Both use two-layer networks (768→384→output) with ReLU and dropout (p=0.2). Total loss:

$$L_{\text{hierarchical}} = L_{\text{sentiment}} + L_{\text{emotion}}.$$

3.2.3 Mixture of Experts (MoE)

Shared DistilBERT encoder produces mask-aware mean-pooled embeddings. Three experts (bidirectional LSTM, CNN with multiple kernels, lightweight transformer) independently produce emotion logits. A gating network computes soft weights $\mathbf{g} = \text{softmax}(\mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \mathbf{h}_{\text{pooled}}))$, and final logits are $\mathbf{z}_{\text{emotion}} = \sum_{i=1}^3 g_i \mathbf{z}_i$. Loss:

$$L_{\text{MoE}} = 0.5 L_{\text{sentiment}} + L_{\text{emotion}}.$$

3.3 Evaluation

We report accuracy, macro F1, and weighted F1 for classification. Calibration is assessed via ECE and MCE over 10 bins. Confusion matrices distinguish within-sentiment from cross-sentiment errors. We visualize emotion classifier weights using UMAP, coloring by sentiment to assess clustering. For MoE, we analyze average gating weights per emotion to identify expert specialization.

4 Results

4.1 Overall Performance

Table 1 presents comprehensive performance on the test set. The flat classifier achieves highest accuracy (57.75%) and weighted F1 (56.55%), marginally outperforming MoE (57.53%, 55.23%) and hierarchical models (56.73%, 55.90%). Macro F1 scores are nearly identical (44%), suggesting similar performance on rare emotions. These results indicate explicit hierarchical modeling provides no clear performance advantage.

Calibration metrics reveal important differences. The flat classifier exhibits substantially better calibration (ECE: 0.0573, MCE: 0.1039) than the hierarchical model (ECE: 0.1174, MCE: 0.1651), whose multi-task training leads to overconfident predictions. The MoE model shows intermediate calibration (ECE: 0.0656, MCE: 0.1167).

Model	Accuracy	Macro F1	Weighted F1	ECE	MCE
Flat Classifier	0.5775	0.4428	0.5655	0.0573	0.1039
Hierarchical	0.5673	0.4370	0.5590	0.1174	0.1651
MoE	0.5753	0.4422	0.5523	0.0656	0.1167

Table 1: Overall classification and calibration performance on the GoEmotions test set.

4.2 Calibration Analysis

Figure 1 shows reliability diagrams for each model. The flat classifier aligns reasonably with perfect calibration, with slight overconfidence in the 0.75-0.85 range. The hierarchical model exhibits severe overconfidence in mid-to-high confidence ranges—predictions at 60-80% confidence achieve only 40-65% accuracy. The MoE model shows moderate overconfidence similar to the flat classifier but with larger deviations at highest confidence. For practical deployment, the hierarchical model would require post-hoc calibration before its confidence scores could be trusted.

4.3 Confusion Patterns

Figure 2 shows normalized confusion matrices. All models exhibit a dominant pattern: frequent misclassification of specific emotions as neutral. Approval, annoyance, disapproval, realization, dis-

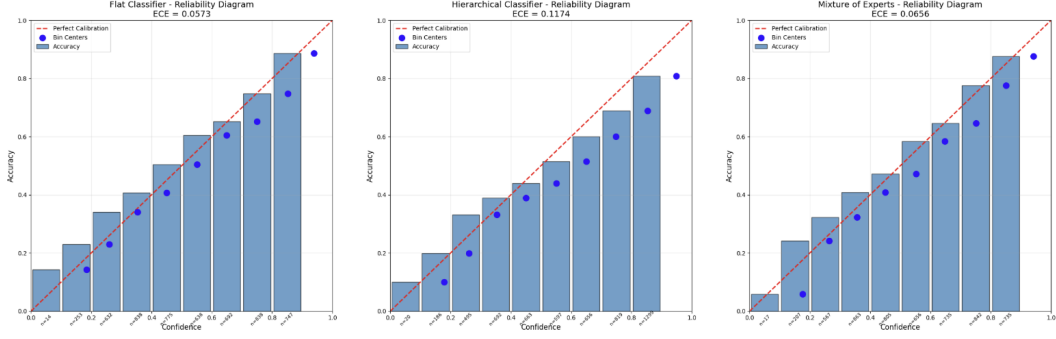


Figure 1: Reliability diagrams for all three models

appointment, caring, curiosity, and anger are often predicted as neutral with high rates, highlighting the inherent ambiguity of neutral expressions.

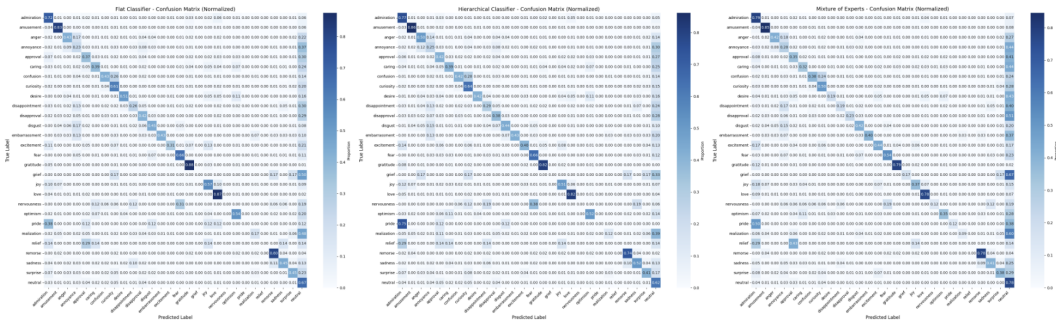


Figure 2: Confusion matrices for all three models (normalized by true label)

Table 2 shows top confusions per model. Direct confusions between non-neutral emotions occur less frequently and typically involve semantically similar pairs (anger/annoyance, confusion/curiosity). Several confusions are asymmetric—realization is frequently misclassified as neutral, but not vice versa—suggesting certain emotions have weaker representations.

Flat Classifier			Hierarchical Classifier			Mixture of Experts		
True → Pred	Count	Rate	True → Pred	Count	Rate	True → Pred	Count	Rate
annoyance → neutral	105	0.367	neutral → approval	88	0.055	approval → neutral	129	0.406
approval → neutral	94	0.296	annoyance → neutral	87	0.304	annoyance → neutral	126	0.441
neutral → curiosity	75	0.047	approval → neutral	85	0.267	disapproval → neutral	113	0.514
neutral → approval	71	0.044	neutral → curiosity	83	0.052	neutral → admiration	68	0.042
neutral → disapproval	67	0.042	neutral → disapproval	72	0.045	curiosity → neutral	66	0.283
disapproval → neutral	63	0.286	disapproval → neutral	61	0.277	realization → neutral	65	0.596
realization → neutral	52	0.477	neutral → annoyance	59	0.037	neutral → curiosity	58	0.036

Table 2: Top 7 Most Common Confusions Across Models

4.4 Learned Representations

We visualize emotion representations using UMAP, coloring by sentiment (positive=green, negative=red, ambiguous=blue).

The flat classifier (Figure 3) learns moderate sentiment-based clustering without explicit supervision. Positive and negative emotions form loose groups, though boundaries are not sharp and ambiguous emotions scatter across the space. This demonstrates implicit sentiment structure learning through DistilBERT.

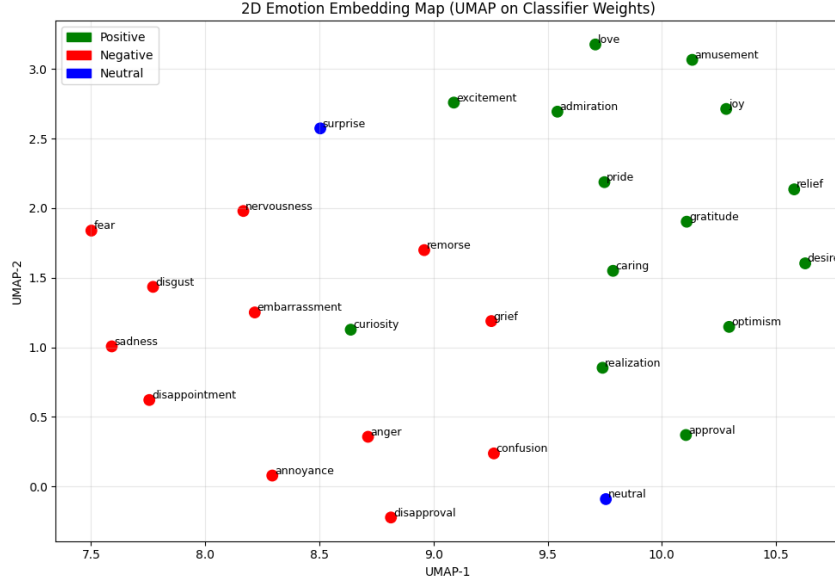


Figure 3: UMAP projection of flat classifier weight vectors

The hierarchical model’s representations (Figure 4) show remarkably similar clustering despite explicit sentiment supervision. This suggests the hierarchical training objective does not fundamentally reshape learned representations, explaining why it does not outperform the flat classifier.

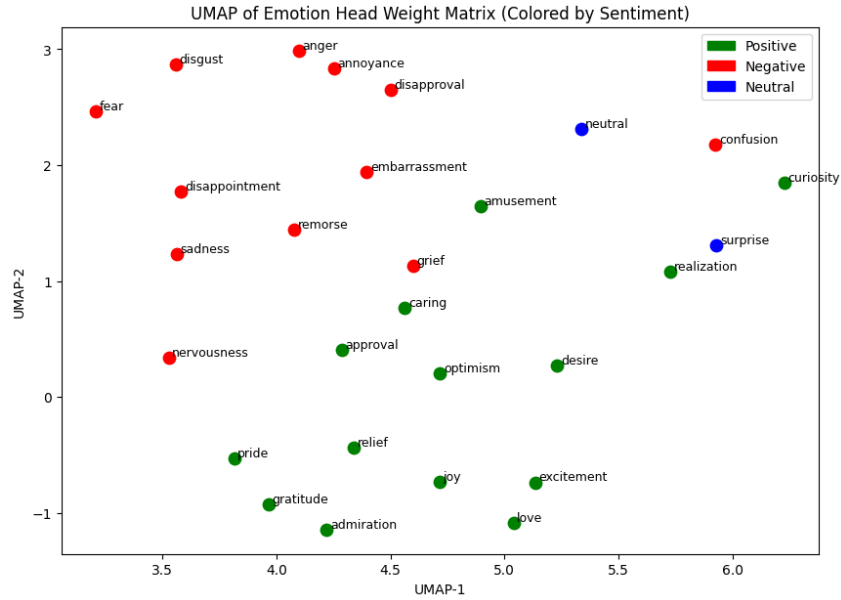


Figure 4: UMAP projection of hierarchical classifier emotion head weights

The MoE gating patterns (Figure 5) reveal strict two-cluster separation loosely aligned with sentiment, indicating the gating mechanism learns hierarchical structure for routing. However, individual experts (Figure 6) show no meaningful sentiment clustering, suggesting specialization occurs through gating rather than expert-level representations.

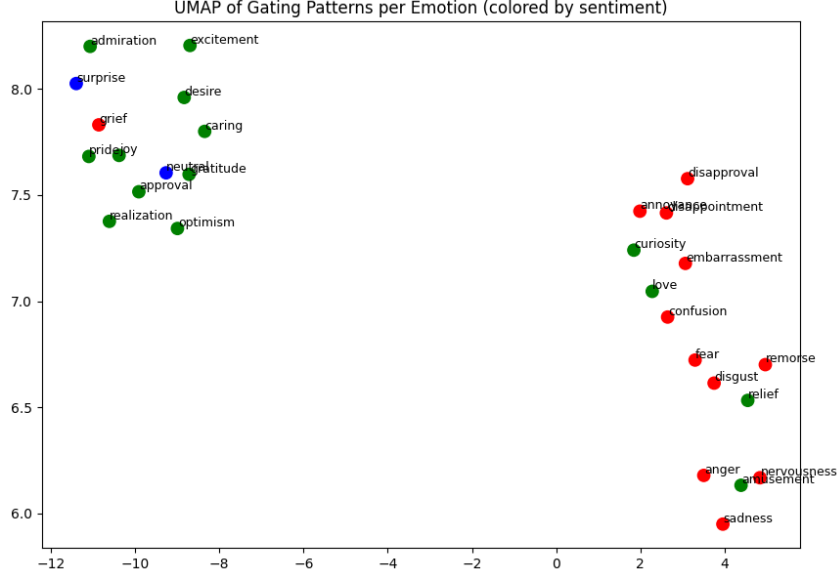


Figure 5: UMAP projection of average gating weights for each emotion

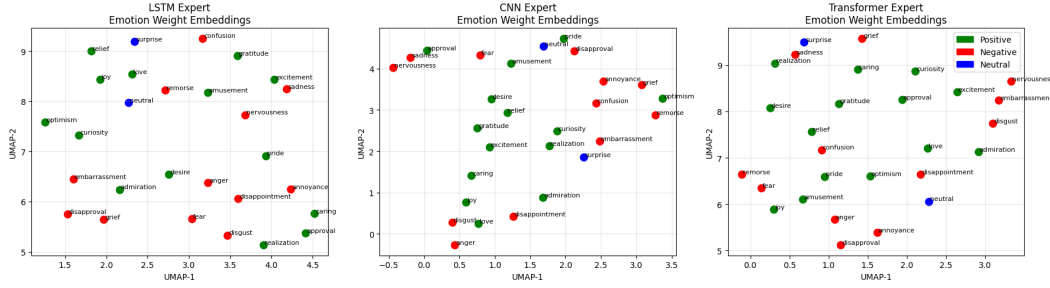


Figure 6: UMAP projections of each expert's weight vectors

5 Discussion

5.1 Key Findings

Our results provide clear answers to both research questions. First, explicit hierarchical modeling does not improve fine-grained emotion classification on GoEmotions. The flat classifier achieves the best performance (57.75% accuracy, 56.55% weighted F1) while the hierarchical model underperforms (56.73%, 55.90%). Macro F1 scores are nearly identical (44%), indicating similar rare-class performance.

However, hierarchical modeling introduces a critical trade-off: severe calibration degradation. The hierarchical classifier's ECE (0.1174) is more than twice that of the flat baseline (0.0573), producing overconfident predictions unsuitable for deployment without post-hoc calibration. Multi-task training appears to create sharper probability distributions that do not reflect true uncertainty.

UMAP visualizations reveal why explicit hierarchy fails to help: the flat classifier already learns sentiment structure implicitly. Both flat and hierarchical models show similar loose sentiment-based clustering, suggesting DistilBERT discovers emotion taxonomy from data alone. Explicit supervision does not meaningfully improve this organization.

Second, the MoE architecture's added complexity is not justified. While achieving intermediate performance (57.53% accuracy), it falls short of the simpler flat baseline. The gating network learns sentiment-aligned routing patterns, but individual experts show no meaningful specialization. This

hierarchical routing does not translate to performance gains, and calibration (ECE: 0.0656) is worse than the flat model.

5.2 Limitations

Several limitations warrant consideration. We simplified multi-label GoEmotions to single-label classification, potentially discarding co-occurrence information. Alternative taxonomies beyond three-way sentiment grouping might yield different results. We evaluated only DistilBERT; larger models may exhibit different hierarchy-learning behaviors. Finally, loss weighting was not extensively tuned—different values might improve hierarchical model performance or calibration.

6 Conclusion

We systematically compared flat, hierarchical, and mixture-of-experts approaches to fine-grained emotion classification. Our key findings are: (1) explicit hierarchy does not improve classification—the flat classifier achieves best accuracy while learning sentiment structure implicitly; (2) hierarchical modeling severely degrades calibration, making confidence estimates unreliable; and (3) MoE complexity is unjustified, as it does not outperform the simpler baseline despite learned routing patterns.

For practitioners, we recommend flat classifiers for emotion classification tasks with similar structure, offering the best combination of performance, calibration, and simplicity. If hierarchical modeling is pursued, post-hoc calibration is essential.

Future work should explore multi-label classification retaining all annotations, alternative hierarchical structures, evaluation on larger models and different architectures, loss weighting strategies balancing performance and calibration, and cross-dataset evaluation to assess generalization.

This work demonstrates that modern pre-trained language models learn rich semantic structure from data alone. Explicit architectural interventions to encode known taxonomies may be unnecessary—and potentially counterproductive—when strong pre-trained representations are available.